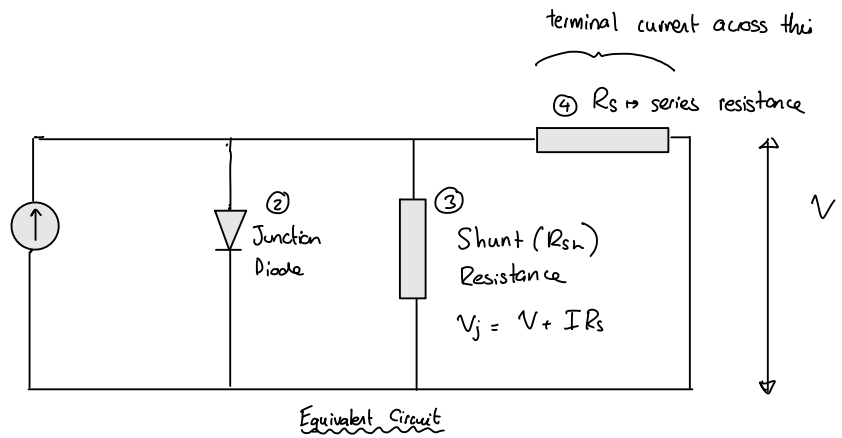


Single Diode Modelling

1) The equivalent circuit

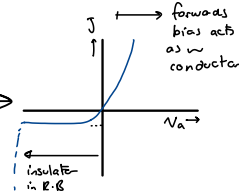
- ① Photocurrent source
 $I_{ph} \propto \text{irradiance}$



- ② Junction diode represents the semiconductor junction & the current associated with carrier injection & recombination.

$$I_D = I_0 \left[\exp\left(\frac{V_j}{nV_T}\right) - 1 \right]$$

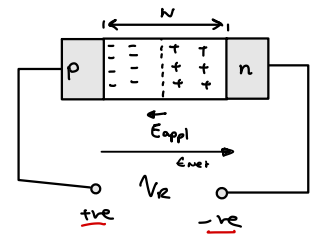
voltage across the junction



When the solar cell develops a voltage, the junction is increasingly forwards biased

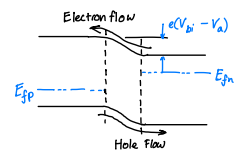
Forward Bias (PN)

$\rightarrow$ reduces built in potential which



$\rightarrow$ the energy hill is reduced

$$\Rightarrow \text{Boltz} \rightarrow \propto \exp\left[\frac{-\Delta E}{kT}\right]$$



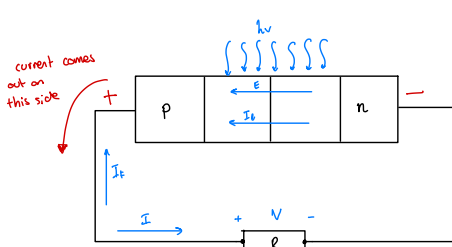
- ③ Shunt resistance $I_{sh} = \frac{V_j}{R_{sh}}$ a large shunt resistance is desirable $\Rightarrow R_{sh} \rightarrow \infty, I_{sh} \rightarrow 0$

& a small R_{sh} allows more current to leak internally instead of reaching the load.

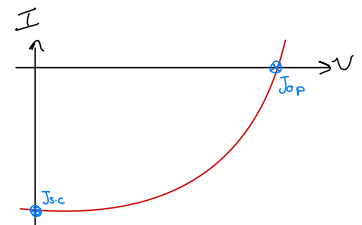
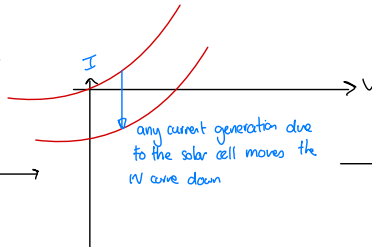
$\rightarrow$ due to: defects / cracks

- ④ $R_s \rightarrow V_{Rs} = IR_s$ represents resistance before current reaches the external circuit i.e. wiring

2) Reminder of underlying mechanism



Under illumination any carriers within the depletion region or one diffusion distance either side will generate a current



1. If $R=0, V=0$: short-circuit conditions

$$\therefore J_{\text{short-circuit}} = J_{\text{optical}}$$

$$J = J_{ph} - J_0 = J_0 \left[\exp\left(\frac{eV}{kT}\right) - 1 \right] - J_0$$

≈ 0 for $V=0$

2. $R=\infty, J=0$ $\Rightarrow$ open-circuit condition

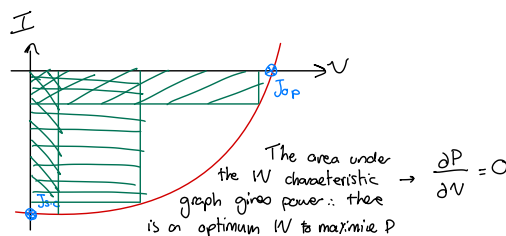
$$\therefore V_{oc} = \frac{kT}{e} \ln\left(\frac{J_{op}}{J_s} + 1\right)$$

The aim is to maximise the power of the solar-cell

$$P = I \cdot V$$

$$= A \left[J_s \left(e^{\frac{eV}{kT}} - 1 \right) - J_{op} \right] \cdot V$$

$J \rightarrow J_A = I$



3) Current Equation

$$I_{ph} \approx q \times \text{absorbed photon rate} \times \text{collection probability}$$

$$I(\text{current out}) = I(\text{photo-generated}) - I_{\text{diode}} - I_{\text{shunt}}$$

$$I_D = I_0 \left[\exp \left(\frac{V_j}{n V_T} - 1 \right) \right]$$

Saturation Current ← I_0

$V_T = \frac{k_B T}{q}$

$$I_s = \frac{V_j}{R_{sh}} \quad V_j = V + I R_s$$

$$\therefore I = I_{ph} - I_0 \left[\exp \left(\frac{V + I R_s}{n V_T} \right) - 1 \right] - \frac{V + I R_s}{R_{sh}}$$

The ideality factor 'n' defines which transport mechanisms dominate

$$n \approx 1 \rightarrow \text{ideal}$$

By considering the idealised case $V_{oc} \rightarrow$ exposed to light but disconnected from an external circuit:

$\therefore R_s \rightarrow 0, R_{sh} \rightarrow \infty \therefore I = 0 \Rightarrow$ no current can leave terminals but recombination still occurs

$$\therefore I_{ph} = I_0 \left[\exp \left(\frac{V_{oc} + I R_s}{n V_T} \right) - 1 \right]$$

$$\therefore V_{oc} = n V_T \ln \left(1 + \frac{I_{ph}}{I_0} \right) \approx n V_T \ln \left(\frac{I_{ph}}{I_0} \right)$$

Fill factor (<https://www.pveducation.org/pvcdrom/solar-cell-operation/fill-factor>)

↳ The ratio of the maximum power from the solar cell to the product of V_{oc} & I_{sc}

$$\therefore FF = \frac{P_{\text{max-power}}}{V_{oc} \times I_{sc}} = \frac{V_{mp} \times I_{mp}}{V_{oc} \times I_{sc}}$$

Implicitness in the equation

$$f(I) = 0 \Rightarrow f(I) = I - I_{ph} + I_0 \left[\exp\left(\frac{V + I R_s}{n V_T} - 1\right) + \frac{V + I R_s}{R_{sh}} \right] \quad \left. \vphantom{f(I)} \right\} \text{ lends itself well to a numerical approach}$$

<https://www.sciencedirect.com/science/article/pii/S0927024803002605?via=ihub>

↑ Exact analytical solutions of the parameters of real solar-cells using Lambert-W function is a 'better' approach
 $\Rightarrow$ num. method always needs to converge.

↳ inverse of $x e^x$

Amit Jain, Avinashi Kapoor,
 Exact analytical solutions of the parameters of real solar cells using Lambert W-function,
 Solar Energy Materials and Solar Cells,
 Volume 81, Issue 2,
 2004,
 Pages 269-277,
 ISSN 0927-0248,
<https://doi.org/10.1016/j.solmat.2003.11.018>.

$$W(\cdot) = \text{lambertw}(\cdot)$$

$$i = \frac{-V}{R_s + R_{sh}} - \text{lambertw} \left(\frac{R_s I_0 R_{sh} \cdot \exp\left(\frac{R_{sh}(R_s I_{ph} + R_s I_0 + V)}{n V_{th}(R_s + R_{sh})}\right)}{n V_{th}(R_s + R_{sh})} \right) \frac{n V_{th}}{R_s} + \frac{R_{sh}(I_0 + I_{ph})}{R_s + R_{sh}}$$

works for $R_s \neq 0$

$$\text{else: } I = I_{ph} - I_0 \left[\exp\left(\frac{V}{n V_T}\right) - 1 \right] - \frac{V}{R_{sh}}$$